Collaborative and parametric insurance on the Ethereum blockchain

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Kick off ANR BLOCKFI

Parametric / Index based Insurance

Insurance Mechanism

Pays a predefined compensation if some measurable quantity reaches a threshold

Examples include

- flight delay
- weather condition

Basis risk

Compensation differs from the actual loss

$\Rightarrow$ Easy to encode in a smart contract

Collaborative insurance

1. Pooling of ressources
   - Participants contributes premiums or funds to a shared pool
2. Decentralized Autonomous Organization
   - Shared governance to set rules and manage the pool

Token

Participants receives token when providing funds

$\Rightarrow$ Token issuance is a common feature of smart contracts

Smart contract

A system associated to

- State variable, e.g. balance of the contract, token exchange rate, token holdings
- Users of the contract ⇒ Definition of roles
- Action taken by users ⇒ Method of the smart contract
- Event to document the actions ⇒ Data Processing

Roles

1. Contract owner : Initialize and update the smart contract parameters
2. Surplus providers : Provide and withdraw from the smart contract
3. Policyholders : Underwrite policies, pay premium and claim compensations

Remark

Part of the premium is used to generate tokens that are redistributed to the policyholder.

Parameters of the smart contract

1. Premium loadings η_1 and η_2

$$\pi = (1 + \eta_1 + \eta_2)\mu,$$

   - μ is the pure premium
   - η_1 loading to reward the token holders
   - η_2 loading given back to the policyholder as tokens

2. Risk levels α_{SCR} and α_{MCR} of solvency capitals

$$\text{SCR} = \text{Quantile}(L - \Pi; \alpha_{\text{SCR}}), \text{ and } MCR = \text{Quantile}(L - \Pi; \alpha_{\text{MCR}})$$

   - L is the liability (stochastic)
   - Π is the premium, yet to be earned (deterministic)

Actions

1. Fund
   - SP locks fund in the smart contract
   - SP receives tokens (given current exchange rate)
2. Withdraw
   - Token holder burns tokens
   - Token holder receives ETH (given current exchange rate)
   - ⚠ Impossible if funds are lower than SCR
3. Underwrite
   - Customer buys coverage
   - Policyholder receives token
   - ⚠ Impossible if funds are lower than SCR
   - ⚠ Value of token decreases
4. Settle
   - Policyholder or contract owner settle a claim
   - Policyholder receives compensation or not
   - ⚠ Value of token increases (premiums earned) or decreases (premiums earned and compensation is paid)
   - ⚠ If funds are lower than MCR (due to compensation) then liquidation

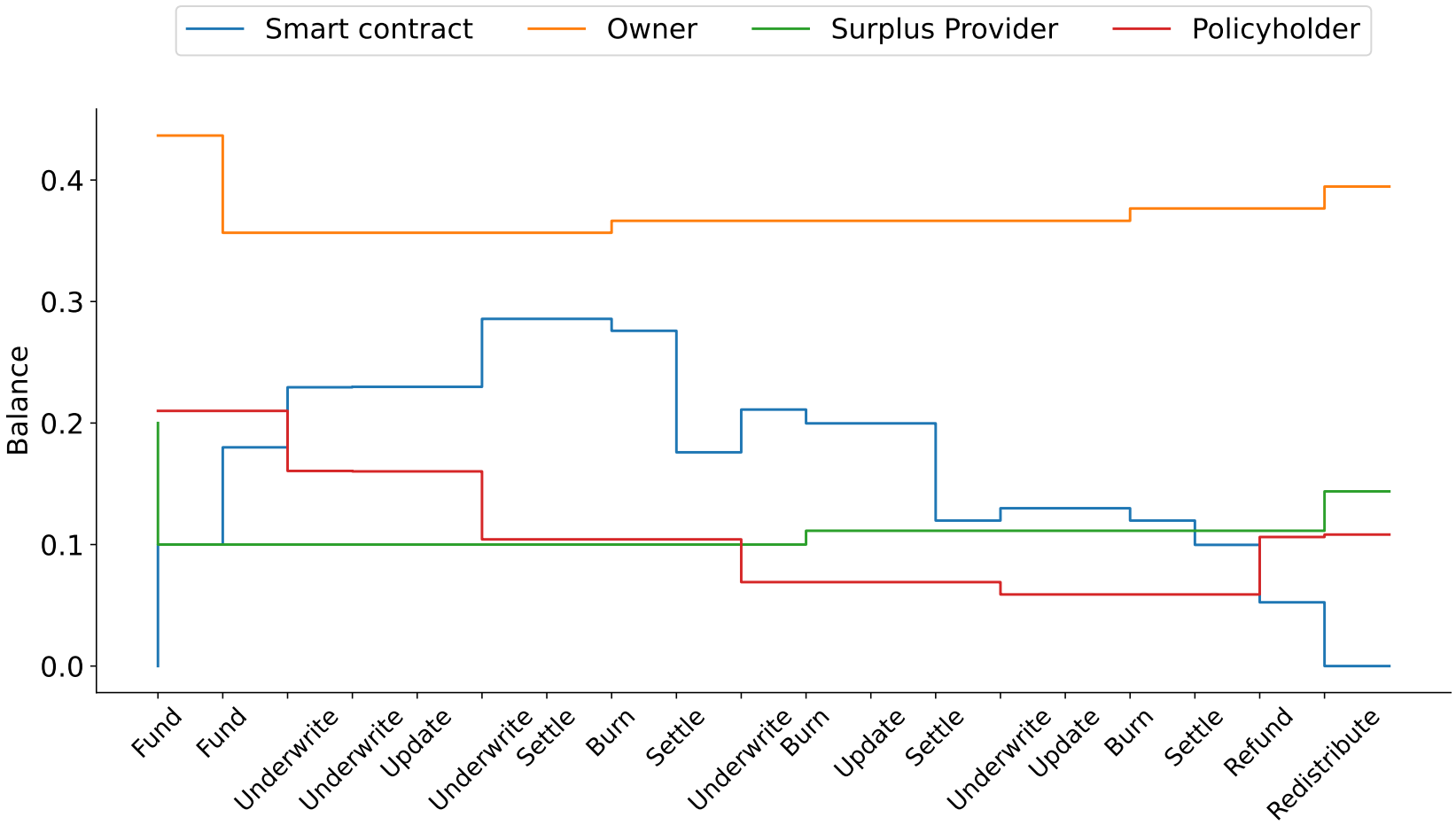

Figure – Balance in ETH of the Ethereum address of the *Owner*, the *Surplus Provider*, the *policyholder*, and the *Smart Contract*.

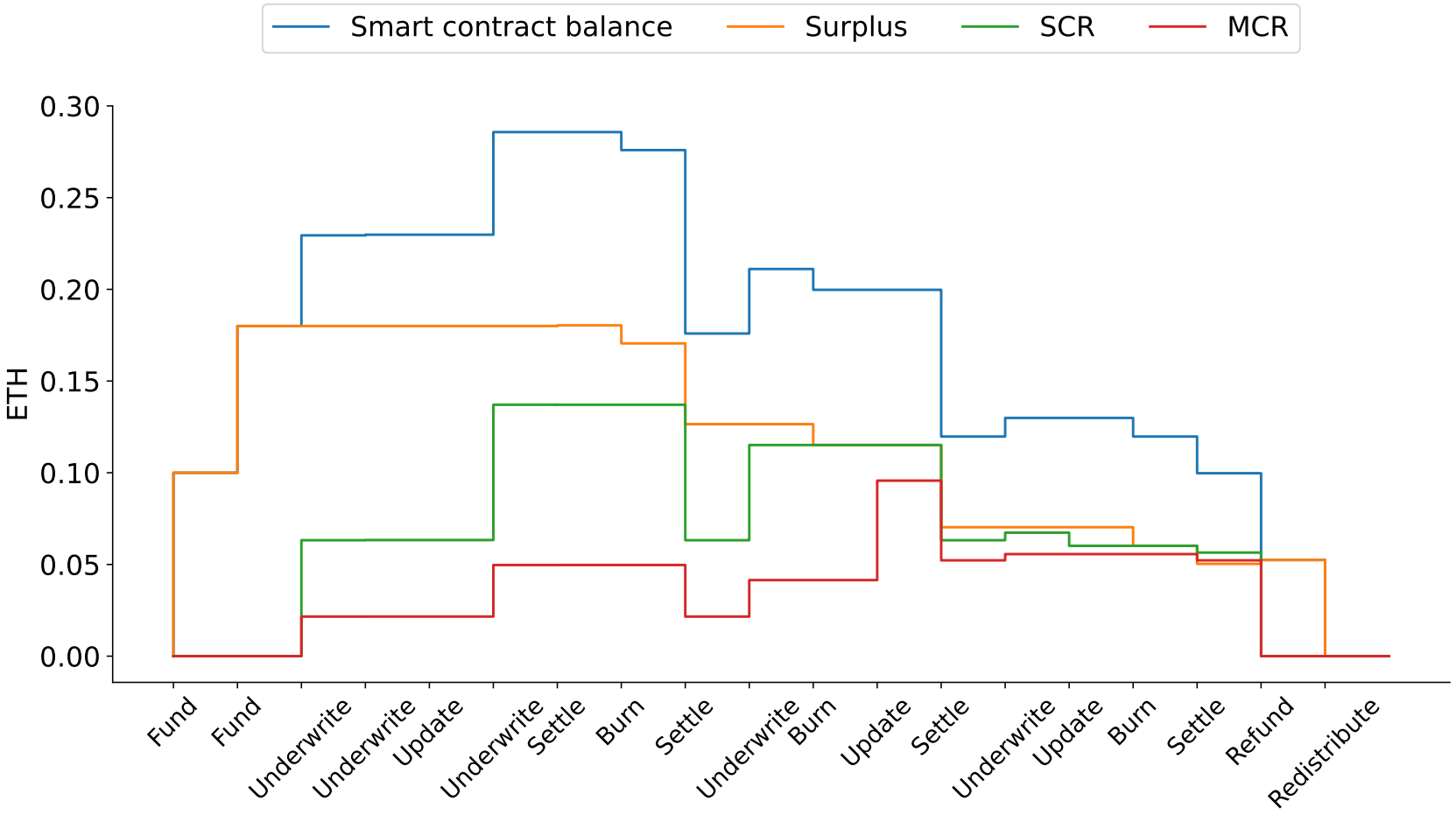

Figure – Trajectories of X, B, SCR and MCR.

Conclusion

- Implementation in solidity
- Deployment of a smart contract on Sepolia testnet
- Data processing in python
- Calculation of the solvency capitals on the fly (despise solidity's limitation)

. https://sepolia.etherscan.io/address/0xa4f3a8567e1d712fc8a53b3a1b6ec76dd42cc10b

. https://github.com/LaGauffre/smart_parametric_insurance/

Limitations

- Simple math models
  - Pricing formula and data oracle
  - Dependent risks (seasonality, event clustering, network effects)
  - Worsening risks (inflation, climate change)
  - Adaptive premium calculation (experience rating)
  - Fluctuation fiat/crypto exchange rate
- Governance system
  - Biaises in risk perceptions
  - Coalitions
  - Behaviors of surplus providers (mass investor lapse)
- Compliance with regulations
  - Insurance regulation (Pillar 2 of Solvency II)
  - Anti-money laundering regulations (KYC)
- Is it insurance ?
  - Policyholder to provide a proof of loss
  - Gambling ?

Math framework

Parametric insurance contract (A_T, I) underwritten at time S

- I is the compensation
- A_T is the triggering event that might occur at time $T > S$

Status of the contract over time :

$$\xi_t = \begin{cases} 0, & \text{if } t < T, \\ 1, & \text{if } t \geq T \text{ and } \overline{A}_T, \\ 2, & \text{if } t \geq T \text{ and } A_T, \end{cases}$$

where

$$0 = \text{"open"}, \quad 1 = \text{"closed and not compensated"}, \quad 2 = \text{"closed and compensated"}.$$

Math framework

- The pure premium associated with a contract (A_T, I) is given by

$$\mu = \theta \cdot I,$$

where $\theta = \mathbb{P}(A_T)$.

- The commercial premium is then given by

$$\pi = (1 + \eta_S) \cdot \mu,$$

where $\eta_S = \eta_S^{(1)} + \eta_S^{(2)}$.

⚠ $\eta_t^{(1)}, \eta_t^{(2)}$ and ξ_t are stochastic processes.

Math framework

- N_t is the number of contract underwritten
- The liability is given by

$$L_0 = 0, \quad L_t = \sum_{i=1}^{N_t} Z_i \mathbb{1}_{S_i \leq t, T_i > t}, \quad \text{for } t \geq 0,$$

  where $Z_i = I_i \mathbb{1}_{A_{T_i}}$ (independent).
- The premium collected (yet to be earned) is given by

$$\Pi_t = \sum_{i=1}^{N_t} \pi_i \mathbb{1}_{S_i \leq t, T_i > t}, \quad \text{for } t \geq 0,$$

- The solvency capitals

$$SCR_t = \text{Quantile}(L_t - \Pi_t; \alpha_{\text{SCR}}), \quad \text{and } MCR_t = \text{Quantile}(L_t - \Pi_t; \alpha_{\text{MCR}}).$$

  Calculated through a normal approximation

Math framework

- Balance of the contract

$$B_0 = 0, \text{ and } B_t = \sum_{i\geq 1} \mathbb{1}_{\tau_i^+ \leq t} x_i^+ - \sum_{i\geq 1} \mathbb{1}_{\tau_i^- \leq t} x_i^- + \sum_{i\geq 1} \mathbb{1}_{S_i \leq t} \pi_i - \sum_{i\geq 1} \mathbb{1}_{T_i \geq t} \mathbb{1}_{A_{T_i}} l_i, \text{ for } t \geq 0.$$

- Surplus of the contract

$$X_0 = 0, \text{ and } X_t = \sum_{i\geq 1} \mathbb{1}_{\tau_i^+ \leq t} x_i^+ - \sum_{i\geq 1} \mathbb{1}_{\tau_i^- \leq t} x_i^- + \sum_{i\geq 1} \mathbb{1}_{T_i \leq t} \pi_i - \sum_{i\geq 1} \mathbb{1}_{T_i \geq t} \mathbb{1}_{A_{T_i}} l_i \text{ for } t \geq 0.$$

- M_t number of token holders and token holding is given by

$$Y_t = \begin{pmatrix} Y_t^{(1)} & \dots & Y_t^{(M_t)} \end{pmatrix}$$

- The exchange rate is given by

$$r_0 = 1, \quad r_t = \frac{X_t}{\overline{Y}_t}, \quad \text{for } t \geq 0.$$

Funding the contract

- Investor transfers x ETH to the contract at time $t > 0$.

$$B_{t+h} = B_t + x, \quad X_{t+h} = X_t + x,$$

  where $h > 0$ is an infinitesimal time lapse.
- If the investor is new, then $M_{t+h} = M_t + 1$; otherwise, $M_{t+h} = M_t$.
- The j-th component ($j \in \{1, \ldots, M_{t+h}\}$) of the vector $(Y_t)_{t \geq 0}$ is updated as :

$$Y_{t+h}^{(j)} = Y_t^{(j)} + y,$$

  where $y = x / r_t$.

Underwrite I

A customer wishes to purchase an insurance contract (A_T, I) at time $t = S$.

- The premium for such a parametric insurance contract is given by :

$$\pi = (1 + \eta_t) \cdot I \cdot \theta,$$

  where $\theta = \mathbb{P}(A_T)$.

- The liability and the cumulative premium collected would then be updated to :

$$\widetilde{L}_t = L_t + I \cdot \mathbb{1}_{A_T}, \quad \widetilde{\Pi}_{t+h} = \Pi_t + \pi.$$

- Update solvency capital requirement :

$$\widetilde{\mathrm{SCR}}_t = \mathrm{Quantile}(\widetilde{L}_t - \widetilde{\Pi}_t; \alpha),$$

  and underwrite the contract only if :

$$X_t \geq \widetilde{\mathrm{SCR}}_t. \tag{1}$$

Underwrite II

- If condition (1) holds, the number of contracts, liability, and premium collected are updated as follows :

$$N_{t+h} = N_t + 1, \quad L_{t+h} = L_t + l_{N_{t+h}} \cdot \mathbb{1}_{A_{T_{N_{t+h}}}}, \quad \Pi_{t+h} = \Pi_t + (1+\eta_t) \cdot l_{N_{t+h}} \cdot \theta_{N_{t+h}}.$$

- The SCR and MCR are also updated as :

$$\mathrm{SCR}_{t+h} = \mathrm{Quantile}(L_{t+h} - \Pi_{t+h}; \alpha_{\mathrm{SCR}}), \quad \mathrm{MCR}_{t+h} = \mathrm{Quantile}(L_{t+h} - \Pi_{t+h}; \alpha_{\mathrm{MCR}}),$$

- We add a component :

$$\xi_{t+h}^{(N_{t+h})} = 0,$$

  to the vector Ξ_t that gathers the statuses of the insurance policies.
- The balance of the contract is updated as

$$B_{t+h} = B_t + (1+\eta_t) \cdot \theta \cdot l,$$

  while the surplus X_t remains unchanged.

Underwrite III

- The policyholder receives :

$$y = \frac{\eta_t^{(2)} \cdot \theta \cdot I}{r_t},$$

  tokens.
- If the policyholder is neither an investor nor a former policyholder, the number of token holders increases as

$$M_{t+h} = M_t + 1,$$

  otherwise, $M_{t+h} = M_t$.
- A component $j \in \{1, \ldots, M_{t+h}\}$ of the vector Y_t is updated as

$$Y_{t+h}^{(j)} = Y_t^{(j)} + y.$$

- The exchange rate becomes

$$r_{t+h} = \frac{X_{t+h}}{\overline{Y}_{t+h}} = \frac{X_t}{\overline{Y}_{t+h} + y}.$$

Withdraw

Token holder $j \in \{1, \ldots, M_t\}$ wishes to burn y tokens at time $t > 0$.

- Token holder receives $x = y \cdot r_t$ ETH.

⚠ The transaction is allowed if

$$y \leq Y_t^{(j)}, \; x \leq B_t, \text{ and } x < X_t - SCR_t. \tag{2}$$

- Provided that (2) is satisfied then we update the contract balance and the surplus as

$$B_{t+h} = B_t - x, \text{ and } X_{t+h} = X_t - x.$$

- Update the token balance of token holder j with

$$Y_{t+h}^{(j)} = Y_t^{(j)} - y.$$

Contract resolution I

At time $t = T_i$, contract i, for some $i = 1, \ldots, N_t$, is settled as it becomes known whether the event A_{T_i} has occurred.

1. A_{T_i} does not occur.
   - The number of contracts, liability, and premiums collected are updated as follows,

     $$N_{t+h} = N_t - 1, \quad L_{t+h} = L_t - l_i, \quad \Pi_{t+h} = \Pi_t - (1 + \eta_{S_i}) \cdot l_i \cdot \theta_i.$$

     Recall that S_i is the time at which contract i was underwritten.
   - The SCR and MCR are also updated as

     $$\mathrm{SCR}_{t+h} = \mathrm{Quantile}(L_{t+h} - \Pi_{t+h}; \alpha_{\mathrm{SCR}}), \quad \mathrm{MCR}_{t+h} = \mathrm{Quantile}(L_{t+h} - \Pi_{t+h}; \alpha_{\mathrm{MCR}}).$$

   - The balance of the contract remains unchanged $B_{t+h} = B_t$, while the surplus is updated as

     $$X_{t+h} = X_t + (1 + \eta_{S_i}) \cdot \theta_i \cdot l_i.$$

Contract resolution II

- The premium for contract i is now considered "earned." The exchange rate is updated as

$$r_{t+h} = \frac{X_{t+h}}{\overline{Y}_{t+h}},$$

where it is noted that the total supply of protocol tokens remains unchanged between t and $t+h$.

2 If A_{T_i} occurred

- the balance, number of contracts, liability, and premiums collected are updated as follows,

$$\widetilde{N}_t = N_t - 1, \quad \widetilde{L}_t = L_t - l_i, \quad \widetilde{\Pi}_t = \Pi_t - (1+\eta_{S_i}) \cdot l_i \cdot \theta_i.$$

- The SCR and MCR are also updated as

$$\widetilde{\mathrm{SCR}}_t = \mathrm{Quantile}(\widetilde{L}_t - \widetilde{\Pi}_t; \alpha_{\mathrm{SCR}}), \quad \widetilde{\mathrm{MCR}}_t = \mathrm{Quantile}(\widetilde{L}_t - \widetilde{\Pi}_t; \alpha_{\mathrm{MCR}}).$$

- The balance and surplus are updated as

$$\widetilde{B}_t = B_t - l_i, \text{ and } \widetilde{X}_t = X_t - l_i + (1+\eta_{S_i}) \cdot \theta_i \cdot l_i.$$

⚠ The use of the tilde indicates that this update is not necessarily final, as the surplus may cross the MCR threshold

$$\widetilde{X}_t \leq \mathrm{MCR}_{t+h}.$$

Liquidation I

The smart contract is reset.

- The premium refunded to the policyholder is the amount paid, excluding the participation loading that granted tokens to the policyholder. That is

$$x = \sum_{i=1}^{\widetilde{N}_t} \mathbb{1}_{\xi_t^{(i)}=0} \left(1 + \eta_{S_i}^{(1)}\right) \cdot \theta_i \cdot I_i.$$

- If $\widetilde{X}_t > x$,
  - the premiums are fully refunded,
  - the remaining wealth, $\widetilde{X}_t - x$, is distributed to the investors according to their token holdings, using the updated exchange rate

$$\widetilde{r}_t = \frac{\widetilde{X}_t - x}{\overline{Y}_t}.$$

  - The smart contract transfers

$$x_j = \widetilde{r}_t \cdot Y_t^{(j)},$$

    to all investors $j = 1, \ldots, M_t$.

- If $\widetilde{X}_t \leq x$,

Liquidation II

  - Policyholders are reimbursed pro rata based on the premiums they paid.

- Smart contract variables are reset to their initial state with

$$B_{t+h} = 0, \quad X_{t+h} = 0, \quad L_{t+h} = 0, \quad \Pi_{t+h} = 0, \quad \mathrm{SCR}_{t+h} = 0 \quad \mathrm{MCR}_{t+h} = 0 \quad r_{t+h} = 1.$$

- The number of contracts and the number of token holders remain unchanged :

$$N_{t+h} = N_t, \quad M_{t+h} = M_t.$$

- Token holdings are reset to zero :

$$Y_{t+h}^{(j)} = 0, \quad \text{for all } j = 1, \ldots, M_{t+h}.$$